

EXPT NO: 02

DATE :

HALF/FULL ADDER

AIM:

To realize half/full adder and half/full subtractor

(i) Using X-OR and basic gates

(ii) Using only NAND gates

APPARATUS REQUIRED:-

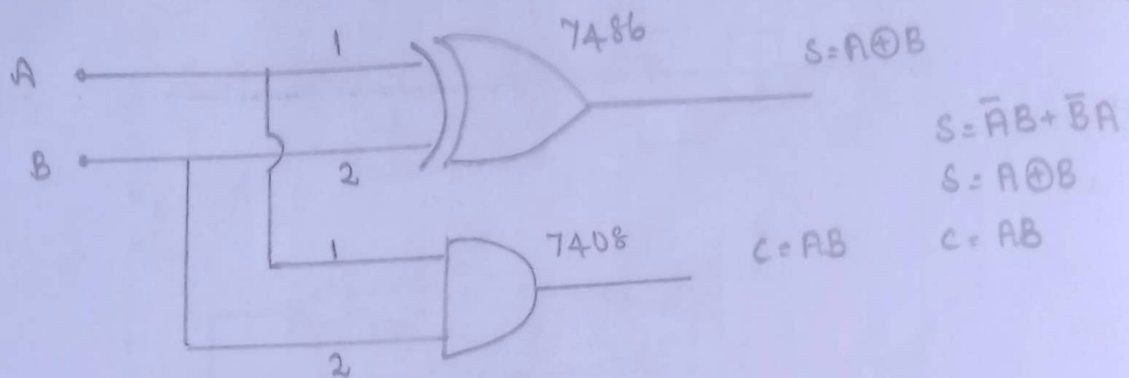
Digital Trainer Kit, IC 7486, IC 7432, IC 7408, IC 7400

and connecting wires.

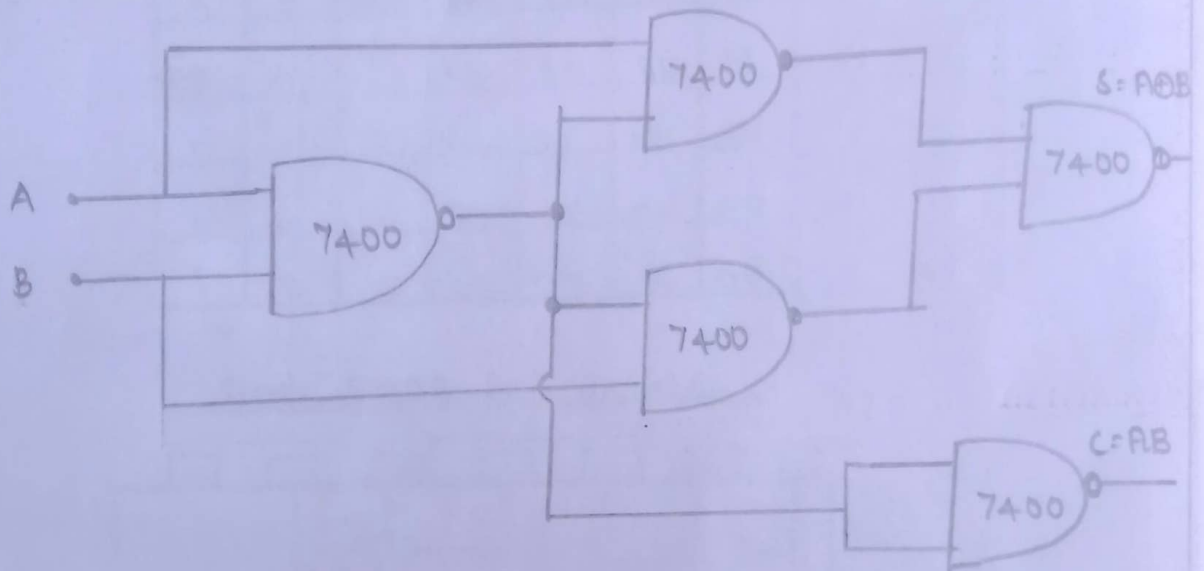
PROCEDURE:-

- 1) Verify the gates.
- 2) Make the connections as per the circuit diagram
- 3) Switch on Vcc and apply various combinations of input according to the truth table
- 4) Note down the output reading of half/full adder and half/full subtractor sum/difference and the carry/borrow bit for different combinations of inputs.

Half Adder using basic gates :-

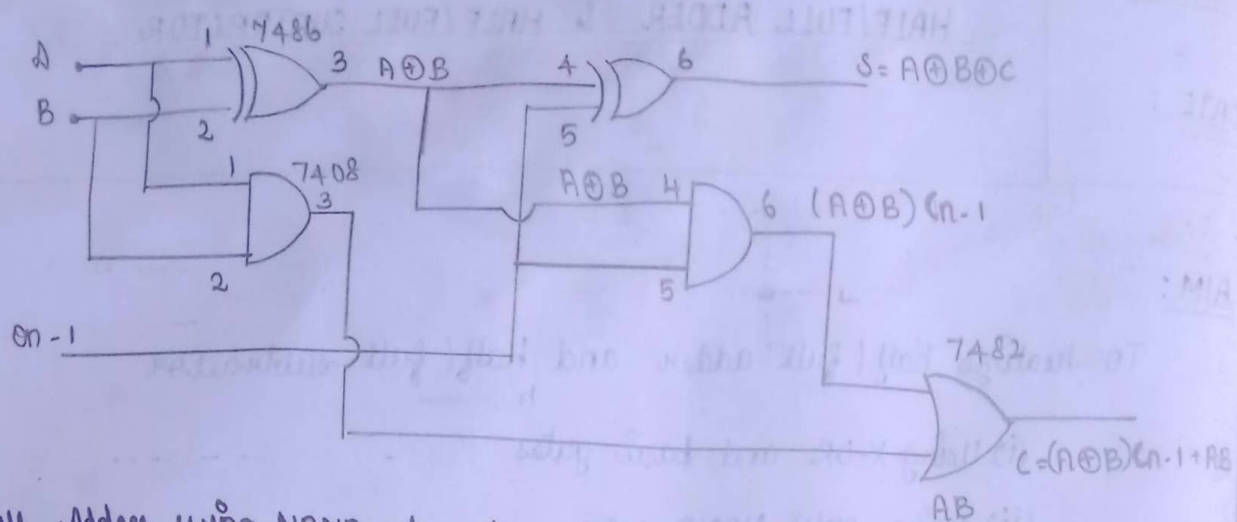


Half Adder using NAND gates only :-

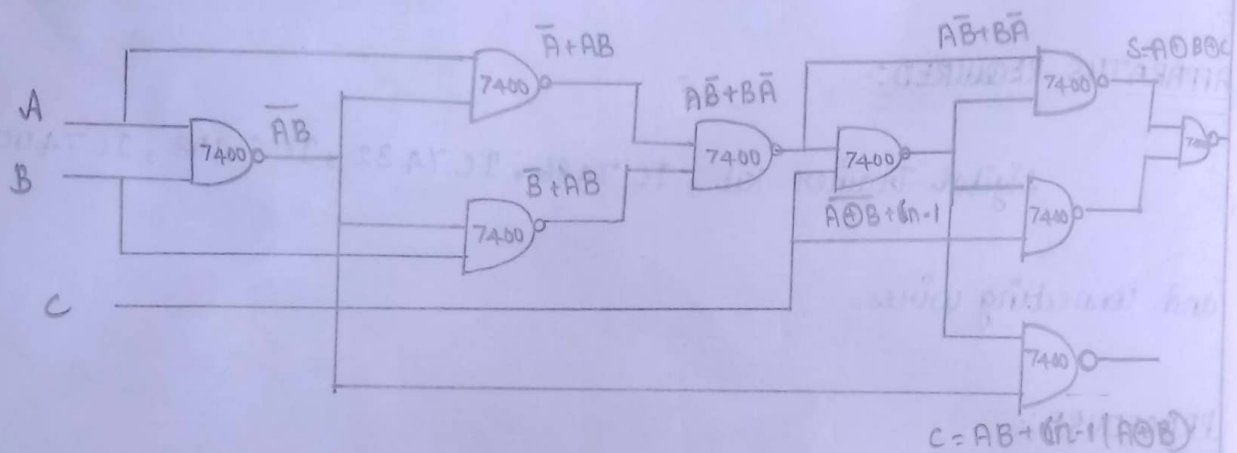


Half Adder					
A	B	S	C	S(v)	C(v)
0	0	0	0	0	0
0	1	1	0	5	0
1	0	1	0	5	0
1	1	0	1	0	5

Full Adder using basic gates:-



Full Adder using NAND gates only:-



Full Adder						
A	B	C _{n-1}	S	C	S(v)	C(v)
0	0	0	0	0	0	0
0	0	1	1	0	5	0
0	1	0	1	0	5	0
0	1	1	0	1	0	5
1	0	0	1	0	5	0
1	0	1	0	1	0	5
1	1	0	0	1	0	5
1	1	1	1	1	5	5

RESULT:-

Thus, the half Adder, full Adder circuits were successfully realized and verified using X-OR and basic gates and NAND gates only. The obtained sum and carry output were found to be in agreement with respective truth tables.